



AGRICULTURE PRO – SMART FARMING ANALYSIS

**DATA ANALYSIS & VISUALIZATION USING EXCEL AND
POWER BI**

PRESENTED BY

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PROJECT OVERVIEW



- **Agriculture data contains information about crops, production, revenue, irrigation and soil.**
- **Managing this data manually can be difficult.**
- **So, I created a project called “Agriculture Pro – Smart Farming Analysis”.**
- **I used Excel and Power BI for this project.**
- **The dashboard helps to analyse and understand agricultural data easily.**



OBJECTIVE



- **To analyse agricultural data easily.**
- **To understand crop production and revenue.**
- **To compare different crops.**
- **To analyse irrigation and water usage.**
- **To understand soil and location details.**
- **To create an interactive dashboard using Power BI.**



TOOLS AND TECHNOLOGIES



- **Microsoft Excel** – Used for data preparation and basic analysis.
- **Power BI** – Used to create the interactive dashboard.
- **DAX** – Used to create calculations and measures.
- **Power BI Slicers** – Used to filter the data.
- **Page Navigation** – Used to move between dashboard pages



DATASET



The dataset contains agricultural information:

- **Crop**
- **Production**
- **Revenue**
- **Irrigation Type**
- **Water Usage**
- **Soil Type**
- **Soil Moisture**
- **Location**
- **Season**



DATA PREPARATION



- **Collected and organized the agricultural data.**
- **Checked the data for errors and missing values.**
- **Prepared the data for analysis.**
- **Used Excel for basic data preparation.**
- **Imported the prepared data into Power BI.**



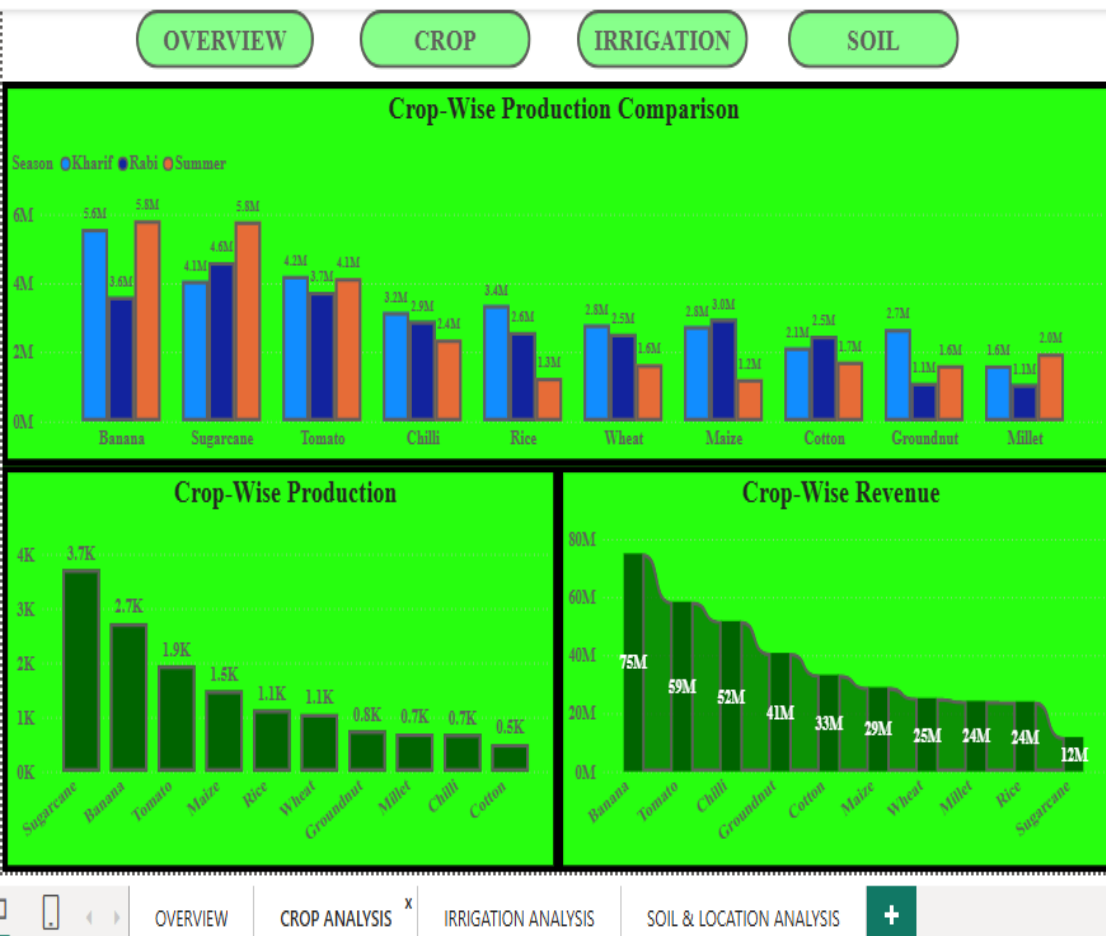
DASHBOARD OVERVIEW



- Interactive dashboard using Power BI
- Different pages for detailed analysis
- Slicers to filter the data
- Charts and cards to show key information
- Navigation buttons to move between pages



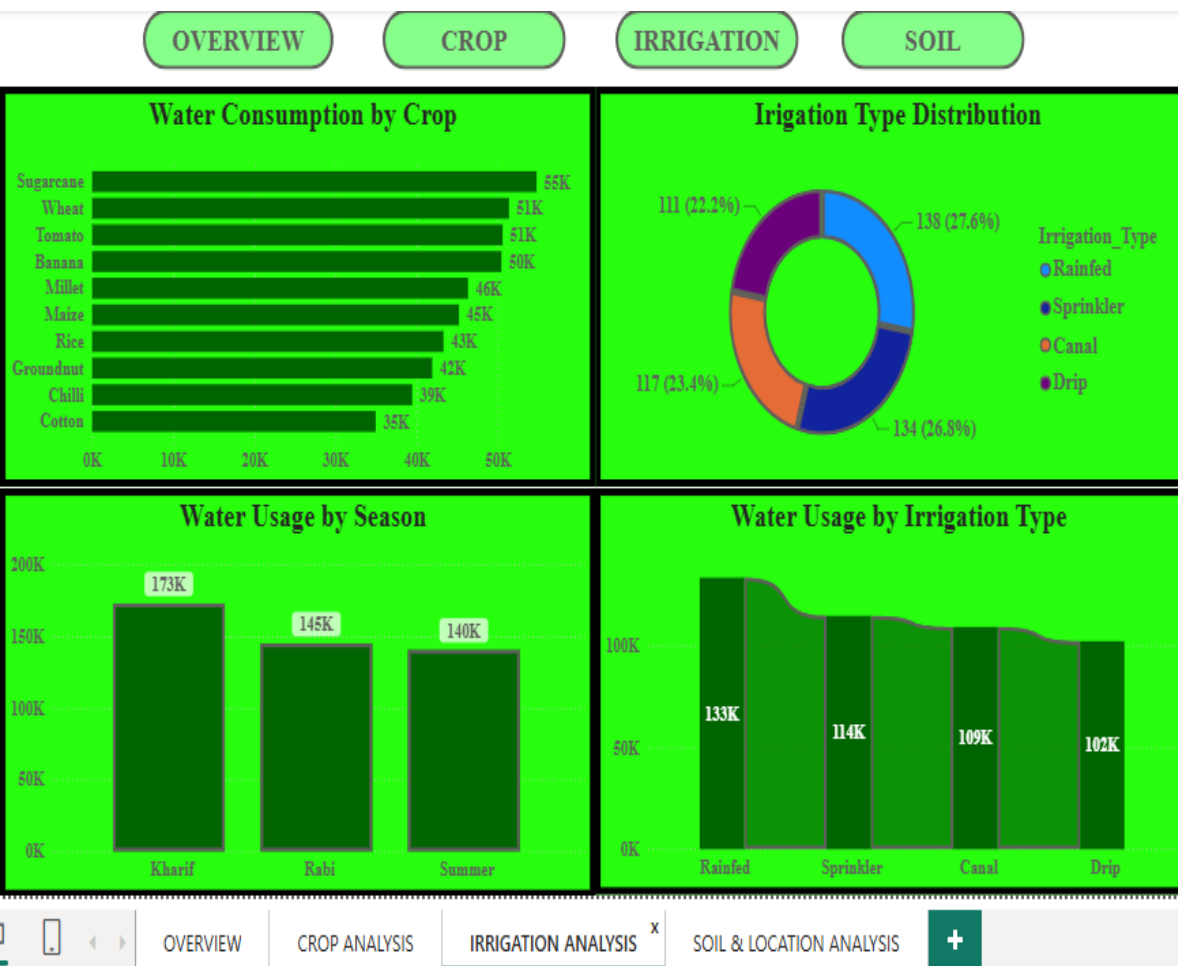
CROP ANALYSIS



- Analyses production of different crops.
- Compares revenue generated by each crop.
- Helps identify the top-performing crop.
- Provides an easy comparison between crops.



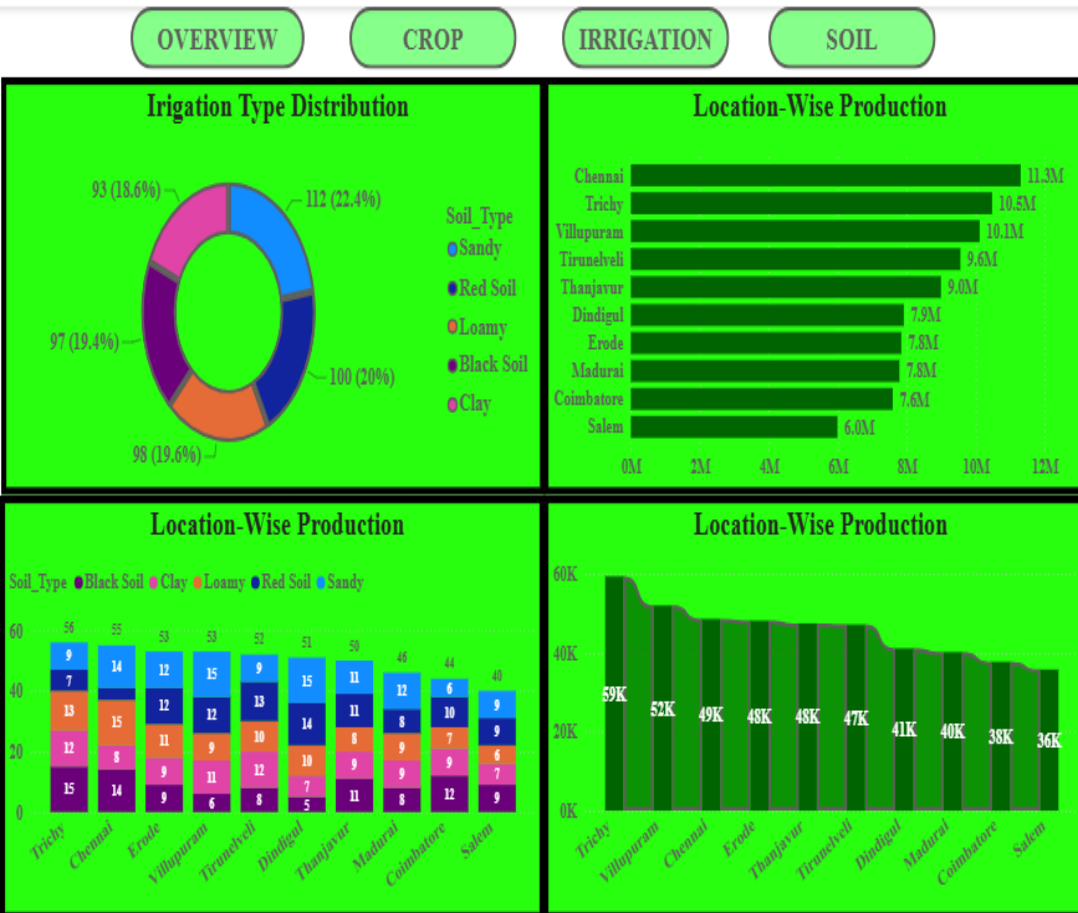
IRRIGATION ANALYSIS



- Analyses different irrigation types.
- Compares water usage and consumption.
- Helps understand water usage across Locations.
- Shows irrigation-related patterns in the data



SOIL & LOCATION ANALYSIS



- Different soil types
- Soil moisture levels
- Production by location
- Season count by location



KEY INSIGHTS



- **Identified the top-performing crops.**
- **Compared crop production and revenue.**
- **Analysed water usage and irrigation efficiency.**
- **Compared soil types and moisture levels.**
- **Identified production differences across locations.**
- **Analysed seasonal patterns by location.**



FUTURE ENHANCEMENTS



- **Add real-time weather data.**
- **Include crop price prediction.**
- **Add rainfall and climate analysis.**
- **Use machine learning for crop yield prediction.**
- **Add more interactive dashboard features**



CONCLUSION



- **Developed an interactive Agriculture Analysis Dashboard.**
- **Analysed crop, irrigation, soil, and location data.**
- **Used Power BI and DAX for data analysis.**
- **Added slicers and page navigation for easy interaction.**
- **Dashboard helps understand agricultural data through visual insights.**



THANK YOU